

Investigator Brochure: GlucoControl-7

1. PHARMACOKINETICS & MECHANISM

GlucoControl-7 is a novel dual-action antidiabetic agent that targets glucose homeostasis through two synergistic pathways.

Mechanism 1: Selective inhibition of Dipeptidyl Peptidase-4 (DPP-4) with 12x higher potency than first-generation inhibitors, prolonging GLP-1 half-life and enhancing glucose-dependent insulin secretion.

Mechanism 2: Activation of AMP-activated protein kinase (AMPK), improving peripheral glucose uptake and hepatic insulin sensitivity without affecting gastric emptying.

2. DOSING & ADMINISTRATION

GlucoControl-7 is administered as a once-daily oral tablet available in 50mg and 100mg doses. Peak plasma concentration (C_{max}) is achieved in 1.8 hours with rapid absorption in the proximal small intestine. Dose escalation: Recommended for patients with HbA1c >9.0% after 8 weeks of initial therapy. No dose adjustment required for mild-to-moderate renal impairment (eGFR >30 mL/min/1.73m²).

3. STORAGE & SHELF LIFE

GlucoControl-7 tablets must be stored in a cool, dry environment at 20°C to 25°C in original amber bottles to prevent moisture absorption. Temperature Warning: Exposure to temperatures exceeding 30°C for prolonged periods (>72 hours) may reduce DPP-4 inhibitory potency by up to 35%, compromising efficacy. Shelf life: 24 months when stored under recommended conditions.